

Digital Evidence Handling & Chain of Custody

Simulated forensic evidence register and integrity workflow

Purpose

Demonstrate a defensible process for identifying, collecting, hashing, storing, transferring, analyzing, and disposing of digital evidence. This is a portfolio example, not a law-enforcement evidence record.

Evidence Handling Principles

- Capture the minimum evidence needed while minimizing unnecessary alteration.
- Record who collected evidence, when, from what source, and why.
- Use cryptographic hashes to support integrity checks for exported files or images.
- Store originals read-only where practical; perform analysis on working copies.
- Document every transfer, analysis action, and disposition.

Evidence ID	Item	Collected by	Collected at	Integrity value	Storage	Status
COC-001	Synthetic EDR export (JSON)	Jamie Christian II	2026-08-20 16:40 ET	SHA-256: 7f3c...2a91 (sample)	Case vault / read-only original	Preserved
COC-002	Synthetic firewall log export	Jamie Christian II	2026-08-20 16:44 ET	SHA-256: 94bd...8f10 (sample)	Case vault / read-only original	Preserved
COC-003	Synthetic suspicious document copy	Jamie Christian II	2026-08-20 16:49 ET	SHA-256: b81e...5d22 (sample)	Restricted malware-evidence folder	Preserved
COC-004	Synthetic memory/host evidence package	Jamie Christian II	2026-08-20 17:02 ET	SHA-256: 6a4f...cc73 (sample)	Encrypted forensic storage	Preserved

Transfer Log

Time	Evidence ID	From	To	Purpose	Integrity check
17:10	COC-001	Collection location	Case vault	Preserve original	Hash matched
17:15	COC-001	Case vault	Working copy	Timeline analysis	Hash matched before copy
17:22	COC-003	Collection location	Restricted storage	Malware-review preservation	Hash matched
Next review	All	Case vault	Retention review	Case closure / retention decision	Re-verify before disposition

Analyst Procedure

- Assign case and evidence identifiers before collection.
- Record date/time with timezone and source system clock context.
- Export/copy evidence using a repeatable method; avoid changing metadata unnecessarily.
- Calculate SHA-256 for collected files when appropriate and record it in the case.
- Place originals in controlled storage and document access.
- Create verified working copies for analysis.
- Log transfers, analysis actions, and any transformation such as decompression or parsing.
- At closure, follow organizational retention/legal-hold requirements and document final disposition.

Important Portfolio Note

The abbreviated hashes shown above are intentionally labeled as samples. A real case would store the complete hash value generated from the actual evidence file. No fake full hash is presented as if it were derived from evidence that does not exist.